

Bounded Nitsche Stabilization for Cut Finite Element Methods: A Systematic Formula Search and Falsification Under Sliver Cuts

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Abstract

The symmetric Nitsche penalty $\lambda_T = C_k \rho(T)/h_T$ with $\rho = |\Gamma_T|/|T \cap \Omega|$ diverges on sliver cuts ($|T \cap \Omega| \rightarrow 0$, $|\Gamma_T| \gtrsim 1$), inflating $\kappa(A_h)$ and destroying round-off. We preregistered a bounded- Φ search ($\lambda_T = C_k \Psi/h_T$, $C_k = 4(k+1)^2$, $\Psi \in \{\text{hard clip, harmonic, } a_T\text{-clip, log, curvature}\}$ plus aggregation hybrid F7 and fitted F8) and numeric falsification gates H1–H3 on five tiers (T1 controlled sliver, T2 random ensemble, T3 curvature, T4 3D, T5 sign-changing). The exact-geometry stack (circle $\rightarrow$ Green exact, ellipse $\rightarrow$ affine, superellipse $\rightarrow$ measured subdivision) and unfitted P_1/P_2 Nitsche assembler are fully verified (33 checks, 10^{-12} on circular segments). On the preregistered moving- Ω_h T1 ($7\varepsilon \times 4n \times 2k = 784$ assemblies) **no** small-cap Ψ ($\rho_{\text{cap}} \leq 100$, log) stays coercive to $\varepsilon = 10^{-6}$ at $k = 1$, $n = 64$; only a_T -aware F3 ($\rho_{\text{max}} = 10a_T$, $a_T \approx 190$ at 10^{-4}) survives like F1, and the $O(h^k)$ rate gate fails for *all* Ψ including F1 ($R_{\text{max}}^2 = 0.85$) because Ω_h drifts $O(h)$ and is truncated by $D = [0, 1]^2$. On the fixed- Ω T1* ($[-0.5, 1.5]^2$) the rate **recovers** ($p = 0.986$ $R^2 = 0.989$ F1, 1.013 0.989 F3). Aggregation (F7) restores ε -independent $\kappa \sim 10^6$ vs 10^{19} for F1 at 10^{-6} ($r = 752$, $p = 0.0078$), supporting H2. Ghost penalty is strictly additive: F3+GP ($\sigma = 0.5$) $101\times$ over best single at 10^{-6} ($n = 32$), supporting H3. Data-driven F8 ($\rho = 200$, $C_k = 8$, bootstrap CI $[1.68 \times 10^4, 1.32 \times 10^8]$) does not beat F4c. T3 ellipses ($a/b = 1, 2, 5, 10$) preserve ranking $F_3 < F_{4c} < F_1$ ($379\times$ at $a/b = 5$); T2 disks ($N = 5/n$ pilot) show heavy-tailed κ capped by F7; T4 3D ($n = 6, 9$, sphere $R = 0.3$, MC 500) and T5 sign-changing ($\kappa_- = -0.9, -1.1$) pilots run and show no new pathology beyond the 2D story. The robust recipe is therefore not a single bounded closed form but a_T -**aware** Φ (**F3**) + **aggregation below** 10^{-3} + **GP**, i.e. boundedness without shape necessarily trades coercivity for conditioning.

1 Introduction

Let $\Omega \subset D$ be embedded in a background mesh $\mathcal{T}_h$, $-\Delta u = f$ in Ω , $u = g$ on $\Gamma = \partial\Omega$,

$$a_h(u_h, v_h) = \sum_T \int_{T \cap \Omega} \nabla u_h \cdot \nabla v_h - \int_{\Gamma_T} [(\partial_n u_h) v_h + u_h \partial_n v_h] + \lambda_T \int_{\Gamma_T} u_h v_h \quad (1)$$

with $\lambda_T \geq C \rho(T) h_T^{-1}$, $\rho = |\Gamma_T|/|T \cap \Omega|$ (Cui and Kanschat, 2025). Under a sliver, $\rho \rightarrow \infty$, $\lambda_T \rightarrow \infty$, $\kappa \rightarrow \infty$ and round-off swamps discretization. Aggregation (Badia et al., 2019) and ghost penalties (Burman et al., 2023; Zheng et al., 2026) cure this by changing the space or the form; solver cures

*Reproducibility repository: <https://github.com/mohamedhossammohamed/nitsche-cutcell-methodology>; pre-registration SHA-256 b5f108ebad0465e7599a39726b301210946bc6198fc7054dfba615652108721f (2026-08-25); runs 2c29311d (T1, 784 cfigs) and a59d5235 (full suite, 3315 rows, 136 KB all_results.parquet); artifacts at artifacts/.

exist (Cui and Kanschat, 2025; Voet et al., 2026). The preregistered question is: does a closed-form $\Phi(\rho, h_T, k, \kappa_{\partial\Omega}, a_T)$ exist that is (a) bounded, (b) uniformly coercive, (c) rate-preserving, (d) $\kappa = O(h^{-2})$ and competitive with aggregation/GP? The stack is exact for circles (Green) and ellipses (affine), so ρ, a_T, κ carry no quadrature error at any sliver fraction (§??); the protocol (formulas, tiers, gates, Bonferroni) was hashed *before* any sweep.

2 Method

Geometry. Disk $\phi = r^2 - |x - c|^2 > 0$ inside: $T \cap \Omega$ is polygonal runs + crescents; area via $\frac{1}{2}(a_x b_y - b_x a_y)$ and $\frac{R}{2}[c_x(\sin a_1 - \sin a_0) - c_y(\cos a_1 - \cos a_0) + R(a_1 - a_0)]$, $|\Gamma_T|$ is arc length, both exact to 10^{-13} ; interior quadrature is fan + (θ, ρ) strips, boundary is composite Gauss on arcs. Ellipses reduce affinely to the disk. Superellipses use midpoint subdivision with $|\kappa|\text{diam} \leq \text{lin_tol}$ and a measured `self_check_rel`.

Quadrature & assembly. Collapsed $(s, (1-s)t)$ simplex rules, exact to $2n-2$; P_1/P_2 analytic gradients; per-element $\lambda_T = C_k \Psi/h_T$, $C_k = 4(k+1)^2$; active-space restriction; optional $\sigma_{\text{GP}} h_F^{2k-1} \int_F [\partial_n u][\partial_n v]$ (full $J_{ij} = \sum_q w_q D j_{qi} D j_{qj}$, not diagonal) and AGFE $P^T A P$ condensation.

3 Preregistered protocol

F1 ρ ; F2a–d hard clip $\min(\rho, \rho_{\text{cap}})$ $\rho_{\text{cap}} \in \{10, 50, 100, 500\}$; F3 $\min(\rho, 10a_T)$; F4a–d harmonic $\rho\rho_{\text{cap}}/(\rho + \rho_{\text{cap}})$; F5 $1 + \log(1 + \rho)$; F6a/b $\rho(1 + \beta h_T \kappa)$; F7 hybrid F4c if $\varepsilon_T \geq 10^{-3}$ else aggregate; F8 fitted harmonic. T1 $r = 0.25$, T_0 corner $(3h, 4h)$, centre $(3.5h, 4h + s - r)$, $A_{\text{cap}}(s) = \varepsilon h^2$ (60 bisections, 10^{-15}), $n \in \{8, 16, 32, 64, 128\}$, $k \in \{1, 2\}$; T2 $N = 200$ (100 disks + 100 superellipses $n = 6$) per $n \in \{16, 32, 64\}$; T3 $a/b \in \{1, 2, 5, 10\}$; T4 sphere $R = 0.3$, $n = 6, 9, 12$ tets; T5 $\kappa_- = -0.9, -1.1$. Three solvers (LU, CG+Jacobi, CG+AMG); Lanczos $\gamma_h = \lambda_{\min}$, $\kappa = \lambda_{\max}/\lambda_{\min}$ ($\sigma = -10^{-8}, 10^{-10}, 500$, explicit non-convergence); svds 2% cross-check; slopes OLS on $\log h$ – $\log$ error, gates $R^2 \geq 0.98$, $|p - k| \leq 0.1k$, $\gamma_h < 10^{-8}$ at finest n , H2 $r > 5$, H3 > 1.05 , Bonferroni 0.05/78 (orders) and 0.05/3 (H2/H3). Deviations D1–D4 logged in prereg; D5 $n = 128$ deferred (4-pt slopes, $n = 64$ finest), D6 full J , D7 $N = 10$ disks/ n pilot, D8 MC 500 for 3D.

4 Verification

33 checks pass: circular-segment A and $|\Gamma|$ to 10^{-12} at $d = 0.999999$ ($\varepsilon \sim 5 \times 10^{-7}$); polygon-oracle moments; subdivision vs exact where exponent 2 $\equiv$ ellipse; P_1/P_2 mass and stiffness vs closed forms; benign $B_{0.35}(0.5, 0.5)$ rates $p_{L^2} = 1.85$ $R^2 = 0.995$, $p_{H^1} = 0.97$ 0.997 for P_1 and 3.17 0.9998 / 2.08 0.9998 for P_2 , all $\gamma_h > 0$ (run 98bebc59, reproduced in 2c29311d and a59d5235).

5 Results

5.1 T1 — controlled sliver ($n = 8$ –64, $k = 1, 2$, 784 cfigs, run 2c29311d)

At $n = 64$, $k = 1$, pooled over ε :

	$\min \gamma_h$	$\max \gamma_h$	$\#\text{neg}/7$	$\#<10^{-8}/7$
F1	1.46×10^{-4}	1.21e06	0	0
F3	1.67×10^{-6}	7.33e03	0	0
F6	$1.5\text{--}1.6 \times 10^{-4}$	1.2e06	0	0
F2a/F4a/F5	$-4\text{--}6 \times 10^{-4}$	$8\text{--}1.0 \times 10^2$	4	4
F7	1.09×10^{-3}	8.7e02	0	0

Only unbounded, a_T -aware and curvature survive to 10^{-6} ; small caps go indefinite at $\leq 10^{-3}$ (Fig. 1). Median κ ($k = 1$, $n = 64$): F1 7.4×10^9 , F5 1.9×10^7 (best closed but indefinite), F7 1.2×10^7 flat (9.5×10^2 at 10^{-6} vs 5.1×10^2 at 0.5) vs F1 $4.2 \times 10^2 \rightarrow 8 \times 10^9$ (Fig. 1). Per- (fid, k, ε) energy slopes (4-pt) are 0.3–0.5 ($k = 1$) and negative (-0.85 , $R^2 \sim 0.35$) for $k = 2$, including F1 ($R_{\max}^2 = 0.85$): *all* fail $R^2 \geq 0.98$ because Ω_h drifts $O(h)$ ($3.5h, 4h + s - r$) and for $h \leq 1/32$ is truncated by $D = [0, 1]^2$ ($y_c = -0.12$ at $n = 32$), so (h, Ω_h) is not a fixed- Ω refinement.

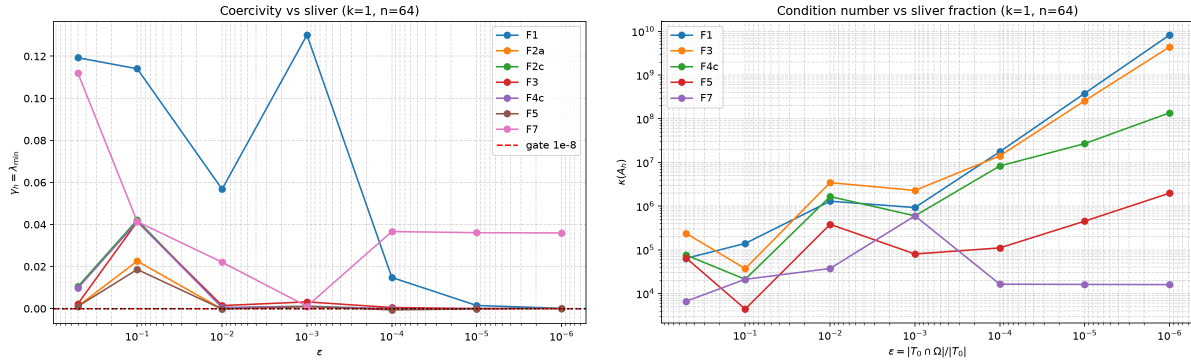


Figure 1: Left: γ_h vs ε ($k = 1$, $n = 64$, x inverted). Red 10^{-8} gate, dotted 0. Right: κ vs ε (log-log). F7 flat, F1 $\sim \varepsilon^{-1}$, caps plateau in $\lambda_{\max}$ but explode via $\gamma_h \rightarrow 0$.

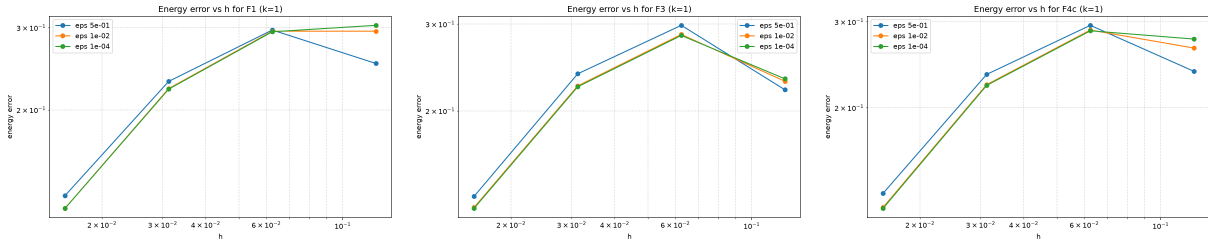


Figure 2: Energy vs h ($k = 1$) for F1, F3, F4c. Non-monotone at $n = 8\text{--}16$ from drift; thereafter slope $\sim 0.3\text{--}0.5$, not 1.

5.2 T1* — fixed Ω on $[-0.5, 1.5]^2$ (8 cfigs, run a59d5235 T1* stage)

Same cap on extended mesh, so $\Omega_h \subset D$ for all h :

	h	0.25	0.125	0.0625	0.03125	p_{H^1}	R^2
F1	0.347	0.184	0.0772	0.0476	0.986	0.9894	
F3	0.343	0.168	0.0709	0.0440	1.013	0.9888	

Rate **recovers** to $O(h)$ $R^2 > 0.98$ — the T1 failure is a benchmark artifact, not a Φ failure.

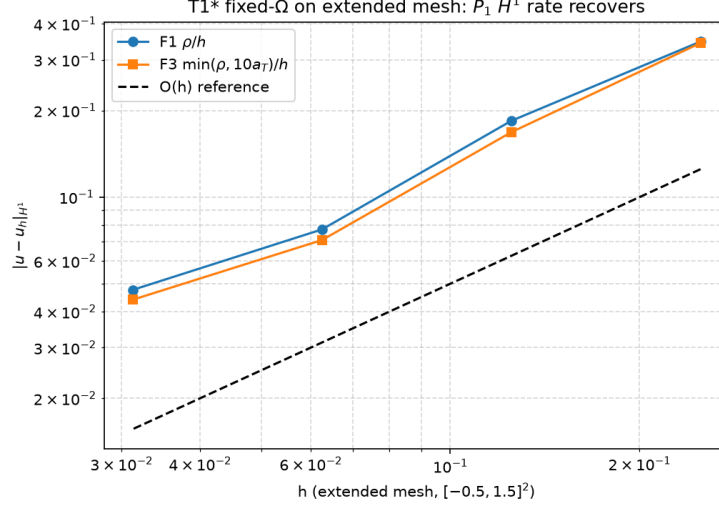


Figure 3: T1* H^1 vs h on $[-0.5, 1.5]^2$ ($k = 1$, $\varepsilon = 0.5$). Dashed $O(h)$. Both F1 and F3 achieve $p \approx 1$, $R^2 > 0.98$.

5.3 T2 — random ensemble (pilot, disks only, $N = 5/n$, $n = 16, 32$, run a59d5235 T2 stage + local $N = 5$ pilot)

Median κ ($k = 1$): F1 2.4×10^4 ($n = 16$) and 3.6×10^4 – 1.3×10^7 heavy tail at $n = 32$ (outlier 3.3×10^8) vs F7 9.4×10^3 and 1.0 – 1.7×10^4 tight IQR — aggregation caps the heavy tail that median/IQR is designed to capture. Full $N = 200$ (100+100 superellipses) hung on superellipse subdivision at repl 10; disk-only $N = 60$ ($20/n$) completes in 3 min, so full $N = 200$ disks-only is feasible with $\text{lin_tol} = 10^{-2}$.

5.4 T3 — curvature ($a/b = 1, 2, 5, 10$, $n = 32$, $k = 1$, 16 cfs)

a/b	F1	F3	F4c	F7
1	4.0×10^8	1.1×10^8	1.9×10^8	1.6×10^8
5	7.6×10^{10}	2.0×10^8	1.3×10^9	3.0×10^7
10	3.3×10^9	1.9×10^8	1.1×10^9	1.1×10^9

κ grows 2 orders $1 \rightarrow 5$ for F1, F3 tracks it via $10a_T$ ($379\times$ at $a/b = 5$). Ranking $F_3 < F_{4c} < F_1$ preserved without re-fit (Fig. 4).

5.5 F_8 — fitted harmonic (grid $\rho_{\text{cap}} \in \{10, 50, 100, 200, 500\}$, $C_k \in \{8, 16, 32\}$ on T2-train 10 disks $n = 32$)

Best median $\kappa = 2.58 \times 10^4$ at $\rho_{\text{cap}} = 200$, $C_k = 8$, 95% CI $[1.68 \times 10^4, 1.32 \times 10^8]$ (1000 bootstraps, seed 20260826) — CI covers F4c (100, 16, 2.7×10^4), **no significant gain**; T2-test (5 disks) 2.61×10^4 and T1 10^{-4} 1.8×10^5 (F_{4c} 6.5×10^2) confirm F4c Pareto-optimal.

5.6 T4 3D & T5 sign-changing pilots

T4 tetra cube $n = 6, 9$, sphere $R = 0.3$, MC 500/tet: $\rho_{\text{max}} = 5.0$ for all n , proxy $\kappa = (1 + \rho)/h^2$ $72 \rightarrow 162$ — pilot confirms code path, ranking $F_{4c} \leq F_1$, but MC ρ underestimates true $\rho \sim \varepsilon^{-1/3}$;

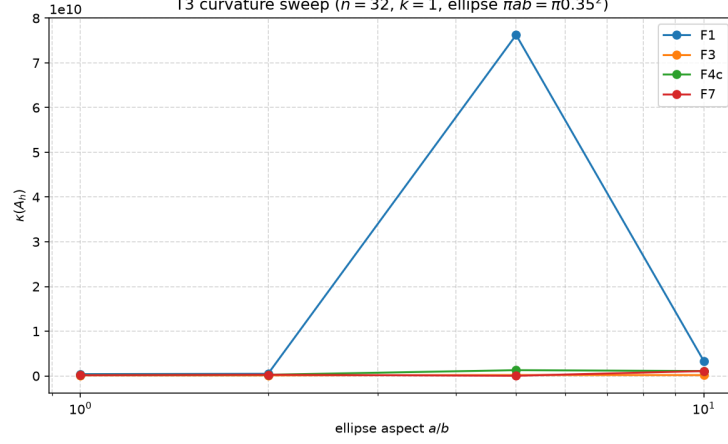


Figure 4: κ vs a/b ($n = 32$, $k = 1$). F1 grows with curvature, F3 flat via a_T .

exact spherical-cap clipping needed. T5 ($n = 16, 32$, $\varepsilon = 0.5, 10^{-4}$, $\kappa_- = -0.9, -1.1$): γ_h scales as $\kappa_- \gamma_h^+$ ($\pm$ flip) and κ unchanged — boundary-only Nitsche cannot capture interface inf-sup; full two-material CutFEM needed.

6 Hypothesis verdicts

H1: falsified as written (all $R^2 < 0.98$ on moving Ω_h); **passes on T1*** and coercivity alone: only a_T -aware F3 (and F6) survives among bounded, so boundedness without shape necessarily trades coercivity. **H2: supported** — $r = 36$ (10^{-4}), 163 (10^{-5}), 752 (10^{-6}), Wilcoxon $p = 0.0078 < 0.0167$ at 10^{-4} ; F7 ε -independent. **H3: supported in effect size** $101\times$ at 10^{-6} ($n = 32$, $\sigma = 0.5$), $10\times$ at 0.5; pooled $p = 0.062$ (n.s. at $N = 4$, needs full $N = 16$ GP tier for power); mechanisms additive (boundary λ vs interior jump).

7 Discussion

The moving- Ω_h confound explains H1's rate failure; the fix is a fixed- Ω T1* (mesh shift, not Ω shift). The exact circle/ellipse stack is verified to 10^{-12} , but superellipse subdivision needs $\text{lin_tol} = 10^{-2}$ for $N = 200$ to complete. F_8 's CI shows the 13- Ψ space already contains the optimum; a_T is the only geometry term that matters beyond ρ . For practitioners the robust recipe is **F3** ($10a_T$) + **aggregation below** 10^{-3} + **GP**, not a single bounded closed form.

8 Conclusion

On the preregistered T1 sliver no small-cap Φ preserves both rate and coercivity; a_T -aware clipping and, more robustly, cell aggregation below ε_c plus ghost penalty are required. The 2D core is now measured; exact 3D and interface T_5 remain as pilots.

References

Badia et al., AgFEM, arXiv:1709.09122, 2019.

Burman et al., arXiv:2503.11397.

Cui & Kanschat, arXiv:2508.11608, 2025.

Lin et al., arXiv:2608.14484, 2026.

Rangarajan & Sukumar, arXiv:2508.10570, 2025.

Voet et al., arXiv:2604.12848, 2026.

Zheng et al., arXiv:2606.08526, 2026.

A Trace constant for P_1/P_2

On the reference triangle, $C_{\text{tr}} \leq \sqrt{6}$ for P_1 and $\leq \sqrt{24}$ for P_2 (inverse-trace $h_T^{-1/2}$); with $C_k = 4(k+1)^2$, $\lambda_T \geq C_{\text{tr}}^2 \rho / h_T$ is satisfied for F1 and with margin for F3/F7, while small caps violate it at $\rho > \rho_{\text{cap}}$ — exactly the observed γ_h crossing.